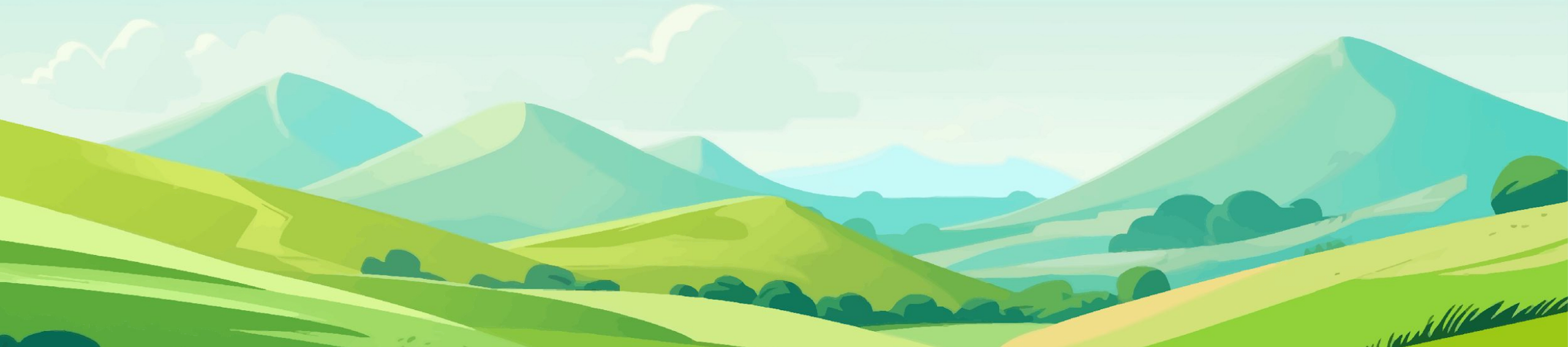
THE LINUX FOUNDATION MEMBER SUMMIT



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Lessons from CNCF KubeStellar's Agentic Workflow Experiments

Andy Anderson, Ashley Wolf, & Nate Waddington

Collaborating on Agentic Workflows

Intro

- Developing a new agentic tool and partnered with CNCF for early feedback
- AI-driven automation that can reason, plan, and execute multi-step tasks
- Written in natural language markdown
- Runs as coding agents in GitHub Actions
- Available at gh.io/gh-aw

How it Works 🔥

- gh aw compile turns markdown into hardened Actions workflows
- Read-only by default, safe-outputs for controlled writes
- Supports Copilot, Claude, Codex as engines

Example: Daily Issues Report

Here's a simple workflow that runs daily to create an upbeat status report:

```
---
on:
  schedule: daily
permissions:
  contents: read
  issues: read
  pull-requests: read
safe-outputs:
  create-issue:
    title-prefix: "[team-status] "
    labels: [report, daily-status]
    close-older-issues: true
---

## Daily Issues Report

Create an upbeat daily status report for the team as a GitHub issue.

## What to include

- Recent repository activity (issues, PRs, discussions, releases, code
- Progress tracking, goal reminders and highlights
- Project status and recommendations
- Actionable next steps for maintainers
```

Use Cases 🌟

- Issue triage
- Repo Assist
- Code review
- Documentation
- Release management

Technical preview

Daily Repo Report - February 9, 2026 #1

github-actions[bot] bot announced in **Daily News**

github-actions[bot] bot yesterday

The gh-aw project is firing on all cylinders! 🚀 With significant commit activity, steady i
team is making excellent progress on improving code quality, debugging capabilities, a

Key Highlights:

- 🎉 **2 new releases** today: v0.41.1 and v0.41.0
- 📊 **1,737 commits** in the last 30 days from **16 contributors**
- 🦎 **100 issues closed** with **190 new issues opened** (net +90 issues)
- 🔀 **50 PRs merged** from **93 opened** PRs
- 🔧 **5 code quality issues** opened today focusing on refactoring opportunities

🔥 Recent Wins

Merged Pull Requests (Last 5):

1. chore: improve orchestration and projects instructions #13666 - Improved orc

CNCF Process and Automation Opportunities

Empowering projects

- CNCF uses the LF AI Guidance Core Principles [1]:
  - Open for Use: AI-generated content is accepted across LF projects.
  - Contributor Ownership: Contributors must ensure AI tool outputs are free of contractual restrictions and provide proper attribution for third-party materials.
  - Project Autonomy: Individual CNCF projects are empowered to develop their own specific recommendations and governance around AI tools.
- CNCF provides tools, partnerships, but delegates AI usage strategies to the projects

[1] Policy Guidance Regarding Use of Generative AI Tools for Open Source Software Development
https://www.linuxfoundation.org/legal/generative-ai

Automation vs. Augmentation

- Recent data [2] identifies AI as a potential "canary in the coal mine" for labor market shifts.
- Two Active Paths:
  - Automative Behaviors: AI directly executes tasks with minimal human involvement (e.g., direct delegation).
  - Augmentative Behaviors: AI enhances human capabilities through iterative collaboration (e.g., refinement and learning).
- Impact on Young Workers [2]
  - Occupations where AI automates labor are linked to a 16% relative employment decline for early-career workers (ages 22-25).
  - Occupations where AI augments labor show stable or continued growth.
- Our Strategy: We are actively investigating Augmentation to increase contributor and maintainer agency

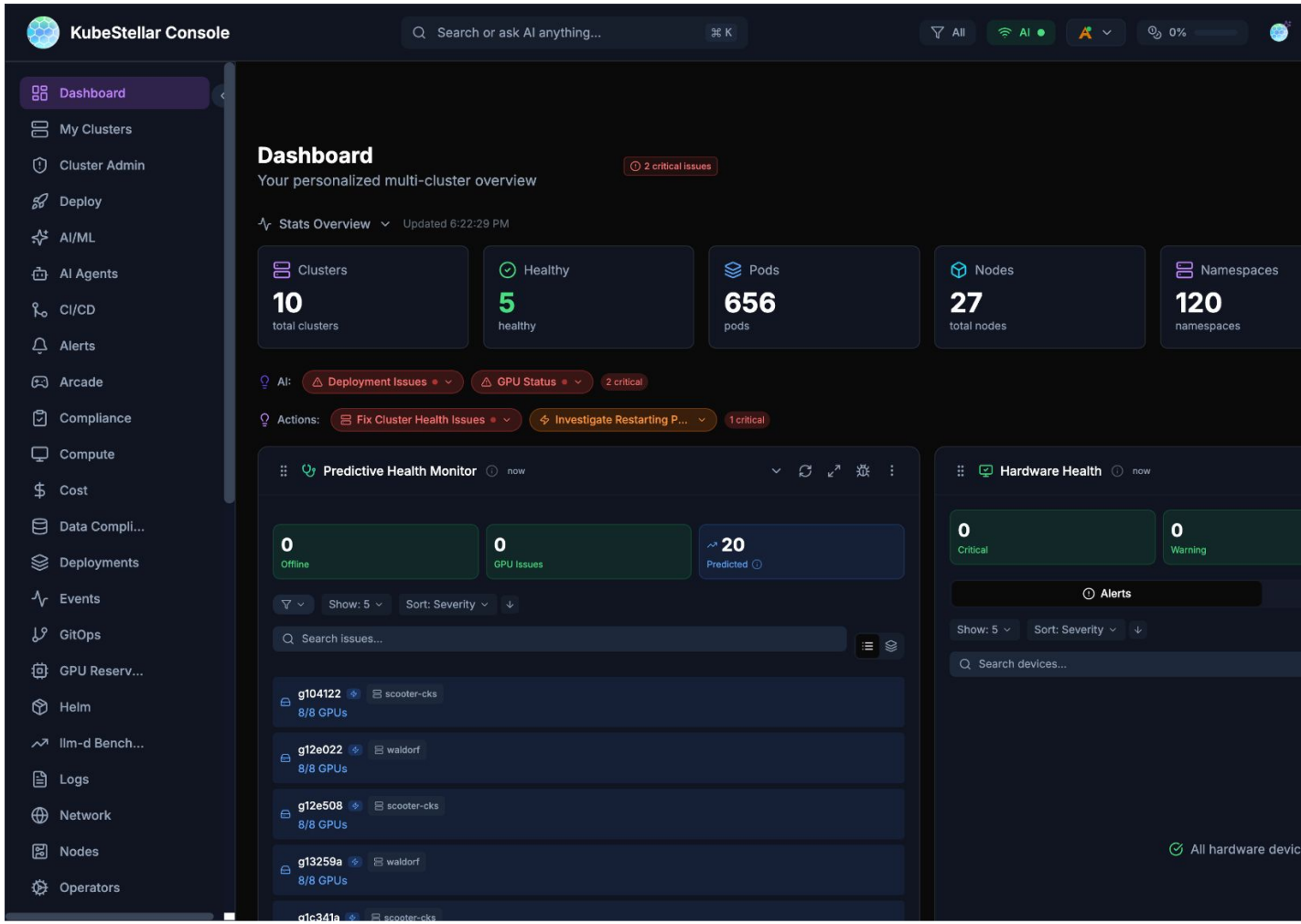
[2] Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence
https://digitaleconomy.stanford.edu/wp-content/uploads/2025/08/Canaries_BrynjolfssonChandarChen.pdf

an Active Lab

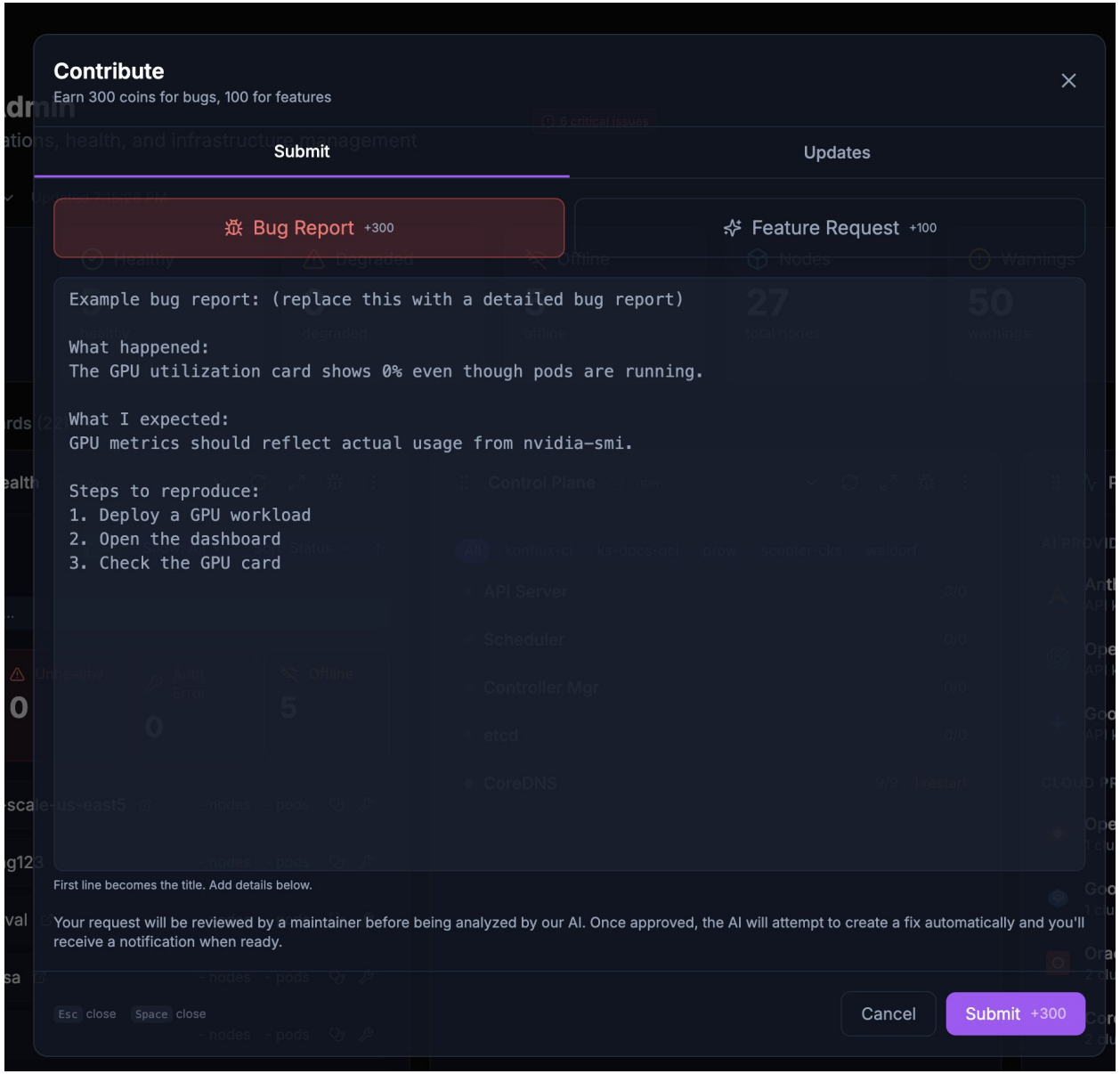
Implementation and Lessons Learned

The Challenge: Scale vs. Quality

- CNCF projects: small maintainer teams, massive community contributions
- KubeStellar Console: 1,300+ PRs in just 6 weeks
- 229 issues filed — impossible to triage manually
- Need: automated quality gates that catch bugs before users do
- Test Driven Design and Implementation is key to success



Quick Demo


Contribute
Earn 300 coins for bugs, 100 for features
Submit
Updates
Bug Report +300
Feature Request +100
Example bug report: (replace this with a detailed bug report)
What happened:
The GPU utilization card shows 0% even though pods are running.
What I expected:
GPU metrics should reflect actual usage from nvidia-smi.
Steps to reproduce:
1. Deploy a GPU workload
2. Open the dashboard
3. Check the GPU card
First line becomes the title. Add details below.
Your request will be reviewed by a maintainer before being analyzed by our AI. Once approved, the AI will attempt to create a fix automatically and you'll receive a notification when ready.
Esc close
Space close
Cancel
Submit +300

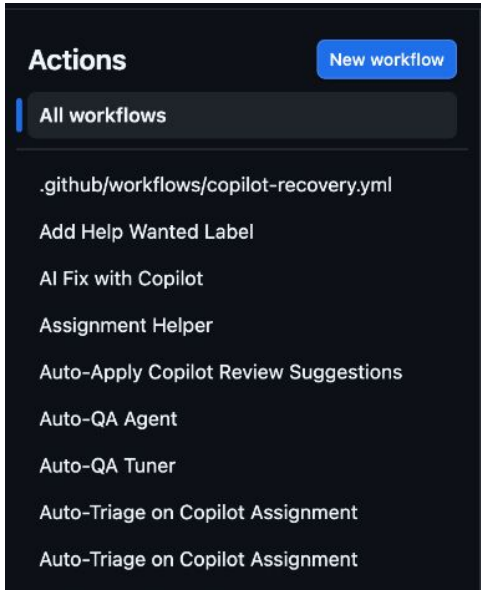
The Auto-QA Pipeline

Detection Phase

- 37 GitHub Actions workflows
- Nightly scans: lint, TS null-safety
- Agentic flows repeatedly on adjusting other Agentic flows
- ‘Goodnight’ agent - update our docs - naturally

Auto-Fix Phase

- Issues auto-created with labels to trigger Copilot
- Copilot assigned automatically
- 193 PRs generated, 101 merged by humans (with or without assist from humans



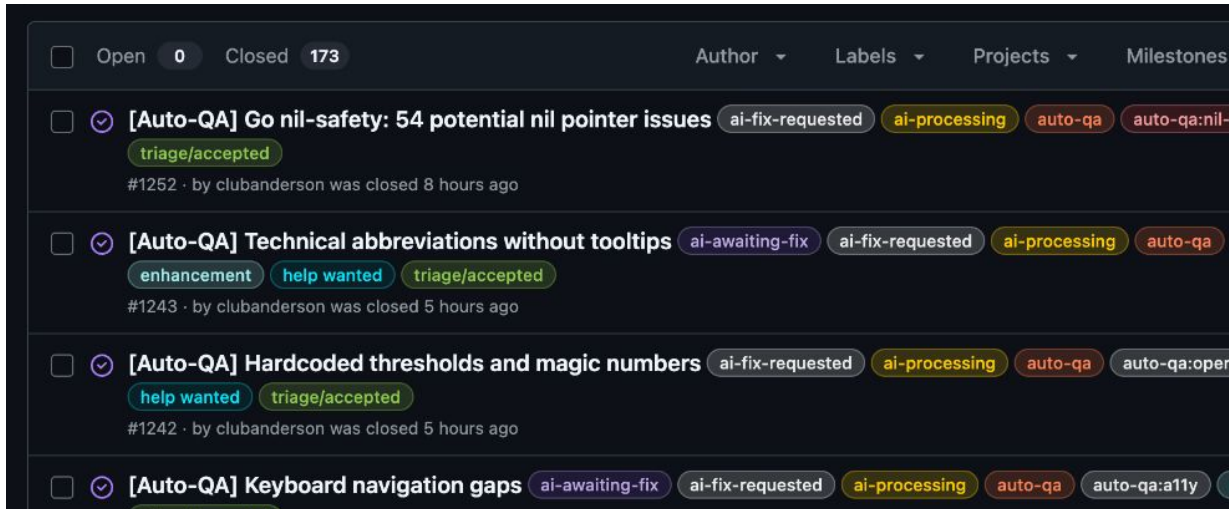
Human Review Phase

- Build + lint CI gates pass first
- Maintainer reviews diff
- Merge via squash with DCO sign-off

The Loop (fully automated)

Bug detected → Issue created → Copilot assigned → PR opened → CI validates → Human merges

67,262 workflow runs in 5 weeks



Results: By the Numbers

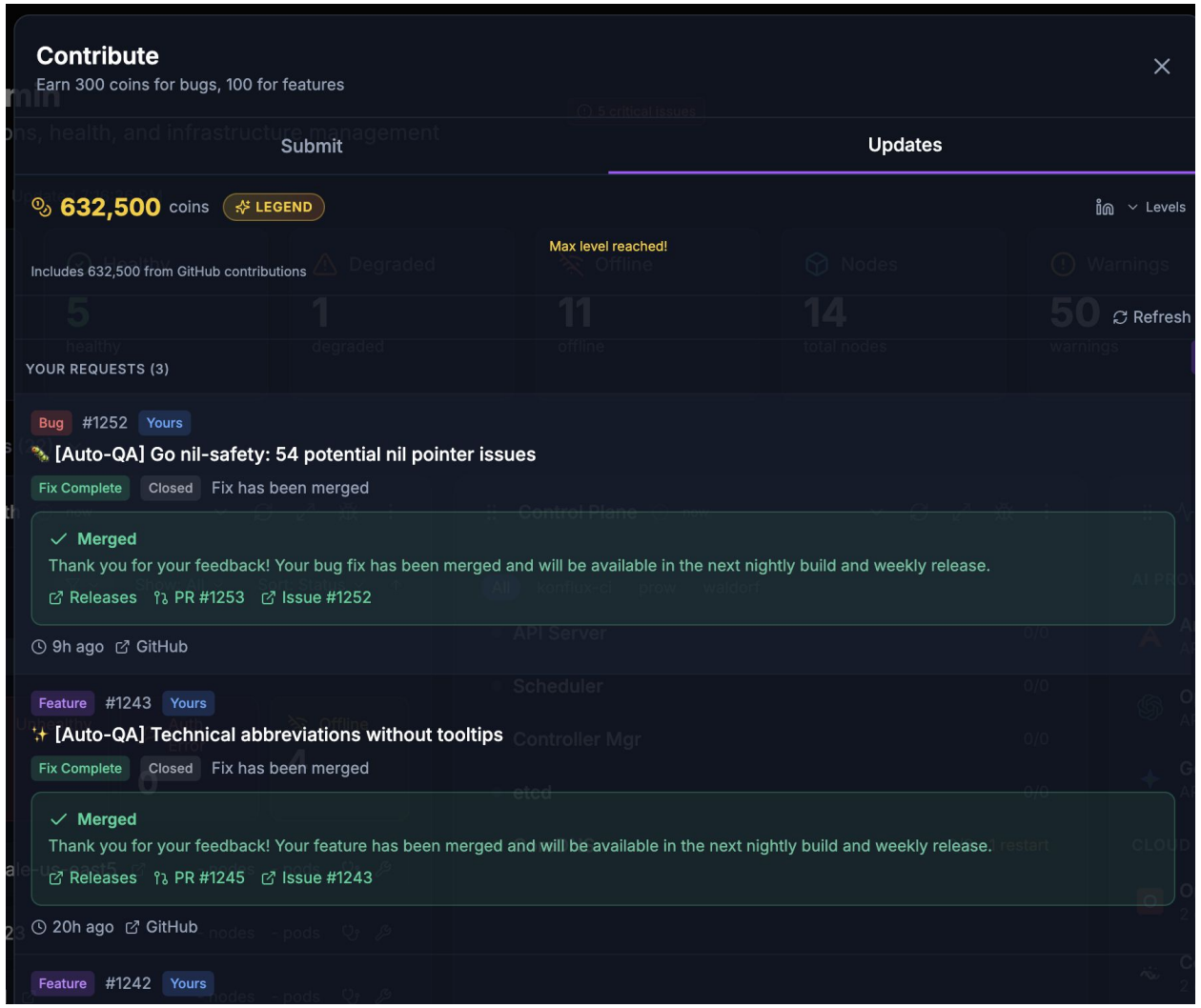
952 PRs merged in 5 weeks (1,302 total opened)

75% of all issues (173/229) auto-detected by workflows

101 Copilot PRs merged — 52% success rate

173 auto-qa issues created and resolved (0 open)

67,262 GitHub Actions workflow runs



Lessons Learned

- Human-in-the-loop is essential (augmentation) — AI generates, humans improve and approve
- 52% Copilot acceptance/approval/merge rate means ~half need human intervention
- Dedupe logic prevents issue spam — same bug = one issue
- Label-driven routing (auto-qa, ai-fix-requested) enables automation
- Gamification drives community bug reporting (300 coins per bug)
- DCO compliance automation is up for debate

How Can Agentic Workflows Benefit You?

Qu

- Scaling to support largest, active projects
- Open source program office automation
  - Org health
- Repo Assist
  - Triage, welcoming new contributors
- Copilot Enterprise for OSS